

Holmium

Ho (Ho I). Strongest classified transitions below 2 μm (NIST): 31 levels, 39 lines.

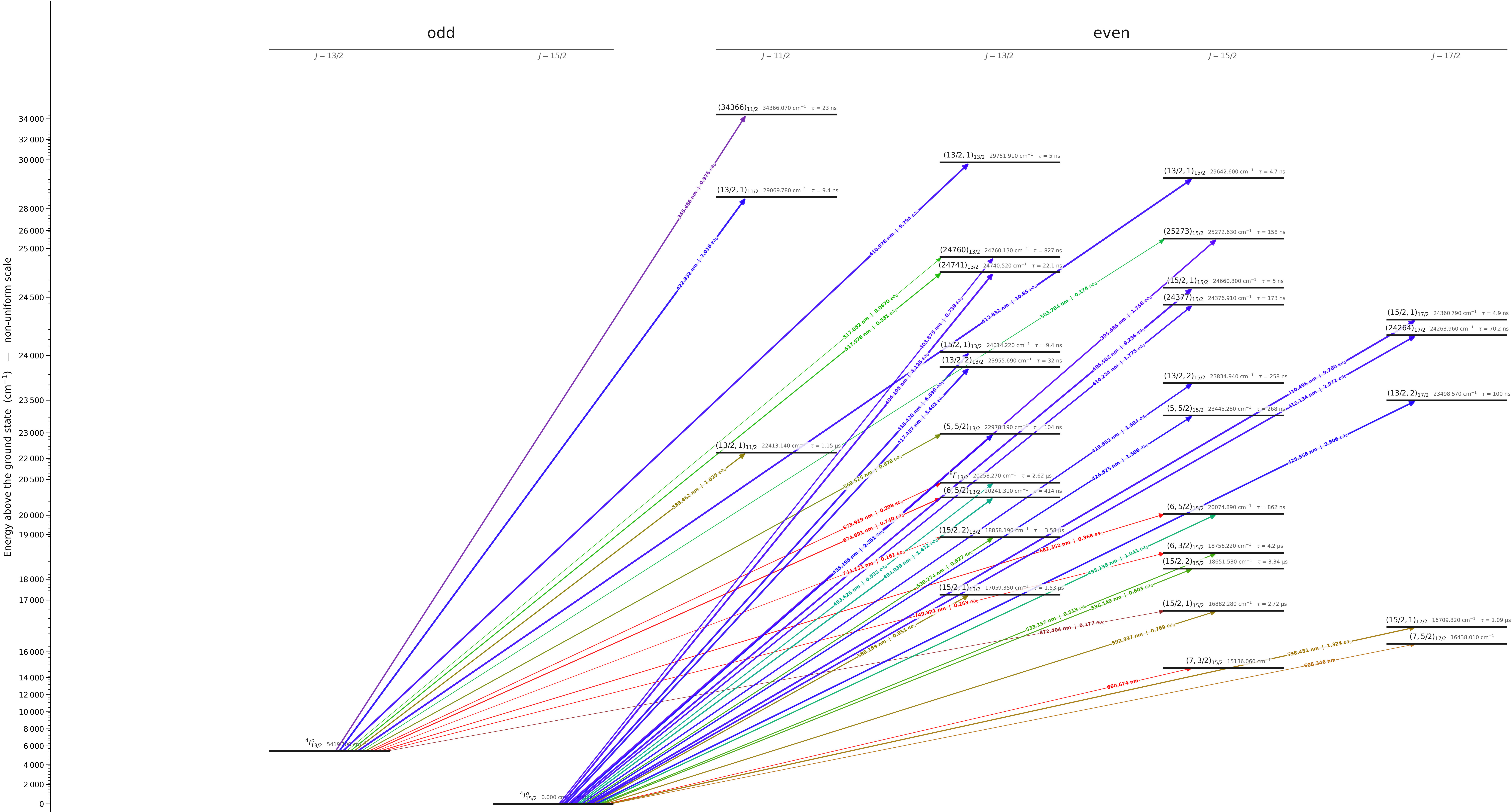
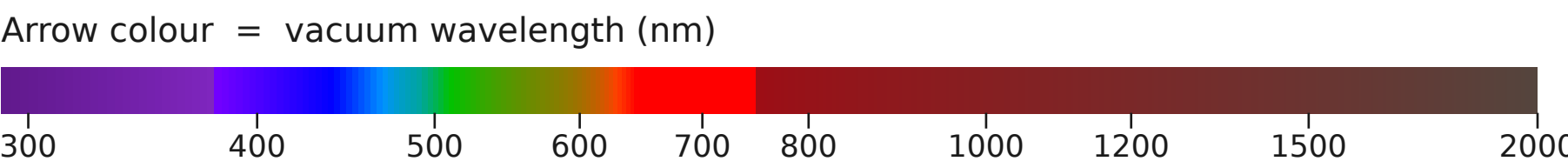
Ho⁺ ionisation limit 48567.00 cm⁻¹ = 6.02154 eV (205.901 nm)

Arrow label: vacuum wavelength | reduced dipole matrix element $|U||er||U'|$ in $e a_0$

Matrix elements are measured or NIST values; * = high-accuracy theory, ≈ = model calculation.

Arrow width grows with the Einstein coefficient A. Dotted: forbidden (clock, M1, E2). Dashed: Rydberg excitation, $n = 70$.

Level caption: energy (NIST, or a measurement of the last 20 years where one exists) and measured lifetime (where one exists).



Key transitions

Transition	λ vac (nm)	λ air (nm)	ν (THz)	⟨U er U'⟩ (e a ₀)	A (s ⁻¹)	Γ/2π up. (MHz)	branch (%)	Use
⁴ f _{13/2} - (34366) _{11/2}	345.4665	345.3675	867.7903	0.976	3.900e+06	6.92	8.97	line from the metastable 4f11 6s2 4lo13/2 level
⁴ f _{15/2} - (25273) _{15/2}	395.6850	395.5730	757.6544	1.756	6.300e+06	1.007	99.5	ground-state line
⁴ f _{15/2} - (24760) _{13/2}	403.8751	403.7610	742.2900	0.739	1.200e+06	0.1924	99.2	ground-state line
⁴ f _{15/2} - (24741) _{13/2}	404.1952	404.0811	741.7021	4.125	3.730e+07	7.202	82.4	ground-state line
⁴ f _{15/2} - (15/2, 1) _{15/2}	405.5018	405.3873	739.3122	9.236	1.620e+08	31.83	81	strong blue line
⁴ f _{15/2} - (24377) _{15/2}	410.2243	410.1085	730.8014	1.775	5.780e+06	0.92	100	ground-state line
⁴ f _{15/2} - (15/2, 1) _{17/2}	410.4957	410.3799	730.3181(3)	9.760	1.550e+08	32.48	75.9	main broad cooling line: MOT, Zeeman slower, imaging
⁴ f _{13/2} - (13/2, 1) _{13/2}	410.9779	410.8619	729.4613	9.794	2.000e+08	31.83	100	line from the metastable 4f11 6s2 4lo13/2 level
⁴ f _{15/2} - (24264) _{17/2}	412.1339	412.0176	727.4152	2.972	1.420e+07	2.267	99.7	cycling line 412.1 nm, two-electron jump
⁴ f _{15/2} - (13/2, 1) _{15/2}	412.8325	412.7161	726.1843	10.85	2.120e+08	33.86	99.6	line from the metastable 4f11 6s2 4lo13/2 level
⁴ f _{15/2} - (15/2, 1) _{13/2}	416.4199	416.3026	719.9282	6.690	8.970e+07	16.93	84.3	strong blue line
⁴ f _{15/2} - (13/2, 2) _{13/2}	417.4374	417.3197	718.1735	3.601	2.580e+07	4.974	82.6	ground-state line
⁴ f _{15/2} - (13/2, 2) _{15/2}	419.5521	419.4339	714.5535	1.504	3.880e+06	0.6169	100	ground-state line
⁴ f _{15/2} - (13/2, 1) _{11/2}	422.8316	422.7125	709.0116	7.018	1.100e+08	16.93	100	line from the metastable 4f11 6s2 4lo13/2 level
⁴ f _{15/2} - (13/2, 2) _{17/2}	425.5578	425.4380	704.4694	2.806	1.150e+07	1.592	100	cycling line 425.6 nm
⁴ f _{15/2} - (5, 5/2) _{15/2}	426.5251	426.4051	702.8718	1.506	3.700e+06	0.5939	99.2	ground-state line
⁴ f _{15/2} - (5, 5/2) _{13/2}	435.1953	435.0730	688.8688	2.251	8.900e+06	1.53	92.6	ground-state line
⁴ f _{15/2} - ⁸ F _{13/2}	493.6256	493.4878	607.3277	0.532	3.400e+05	0.06075	89.1	ground-state line
⁴ f _{15/2} - (6, 5/2) _{13/2}	494.0392	493.9013	606.8192	1.472	2.600e+06	0.3844	100	ground-state line
⁴ f _{15/2} - (6, 5/2) _{15/2}	498.1347	497.9958	601.8301	1.041	1.110e+06	0.1846	95.7	ground-state line
⁴ f _{15/2} - (25273) _{15/2}	503.7040	503.5636	595.1759	0.174	2.990e+04	1.007	0.47	line from the metastable 4f11 6s2 4lo13/2 level
⁴ f _{13/2} - (24760) _{13/2}	517.0516	516.9076	579.8115	0.067	4.700e+03	0.1924	0.39	line from the metastable 4f11 6s2 4lo13/2 level
⁴ f _{13/2} - (24741) _{13/2}	517.5764	517.4323	579.2236	0.581	3.520e+05	7.202	0.78	line from the metastable 4f11 6s2 4lo13/2 level
⁴ f _{15/2} - (15/2, 2) _{13/2}	530.2736	530.1261	565.3543	0.527	2.700e+05	0.04446	96.7	ground-state line

λ, ν from NIST level energies unless a measured frequency for this isotope is available. A: measured value or NIST.

Γ/2π: natural linewidth of the upper level, 1/(2τ).

¹⁶⁵Ho hyperfine structure (I = 7/2)

Level	A (MHz)	B (MHz)	Shift of each F level from the centre of gravity (MHz)
⁴ f _{15/2}	800.583645(6)	-1668.0053(3)	F=4: -24424.99 F=5: -20115.71 F=6: -15019.45 F=7: -9177.08 F=8: -2636.25 F=9: +4548.58 F=10: +12316.13 F=11: +20598.32
(15/2, 1) _{17/2}	654.9(3)	-620(20)	F=5: -21991.97 F=6: -17945.34 F=7: -13253.91 F=8: -7931.35 F=9: -1993.29 F=10: +4542.69 F=11: +11657.04 F=12: +19328.28

Measured A, B constants; sources in the data file.